

Why co-evolve?

Offline coverage is finite. No offline set can span every pulse that iterative search will visit.

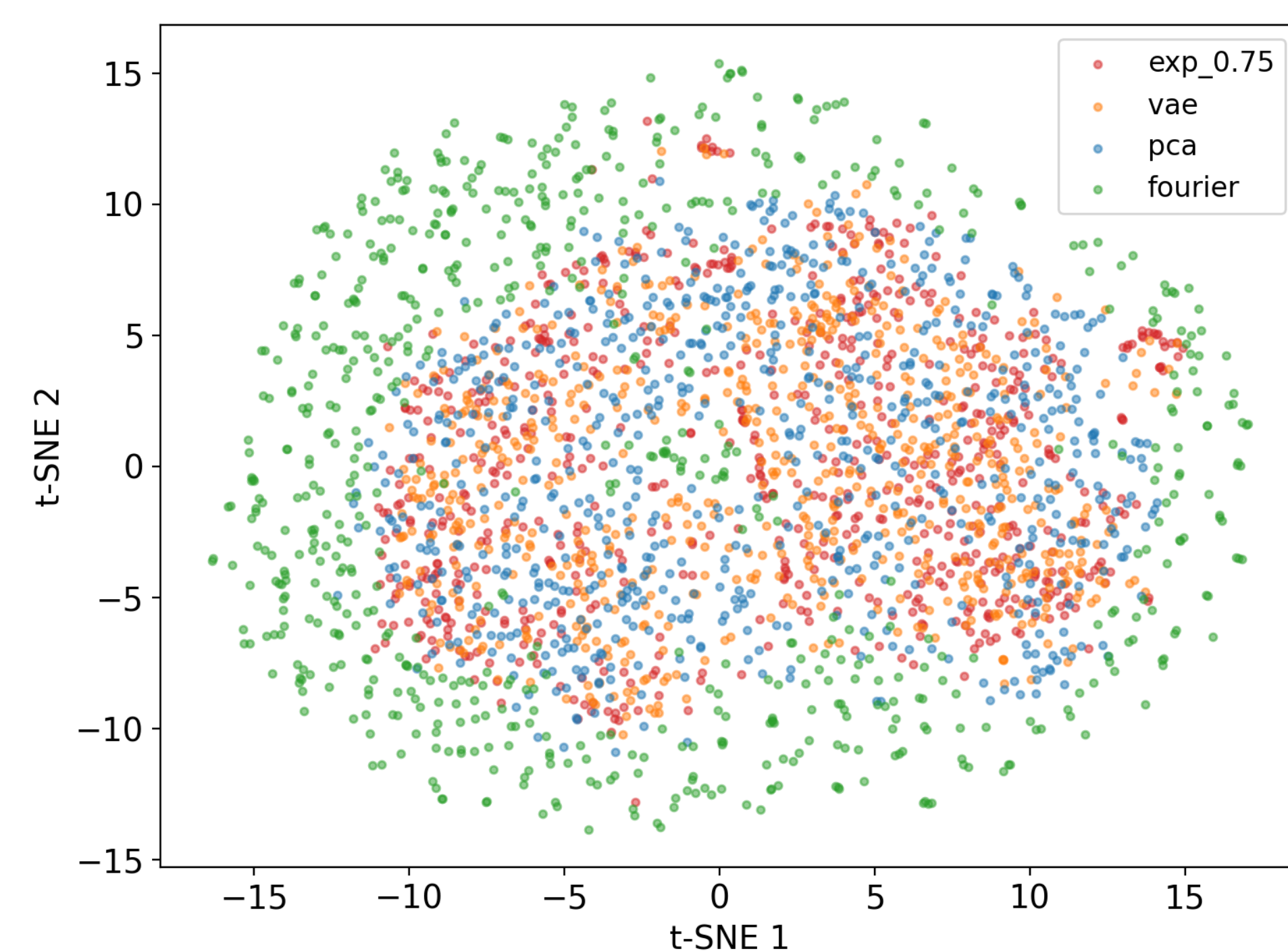
PPO moves the query distribution. Maximizing predicted Lawson reward pushes pulses beyond $\mathcal{D}_0$.

OOD error becomes reward hacking. A frozen surrogate can rank overestimated or non-physical trajectories as optimal.

MULTI-only PPO is impractical. Calling the high-fidelity solver at every update removes the speed advantage.

Co-evolve: fast PPO on the surrogate + budgeted MULTI relabeling of current-policy pulses.

MULTI-IFE coverage



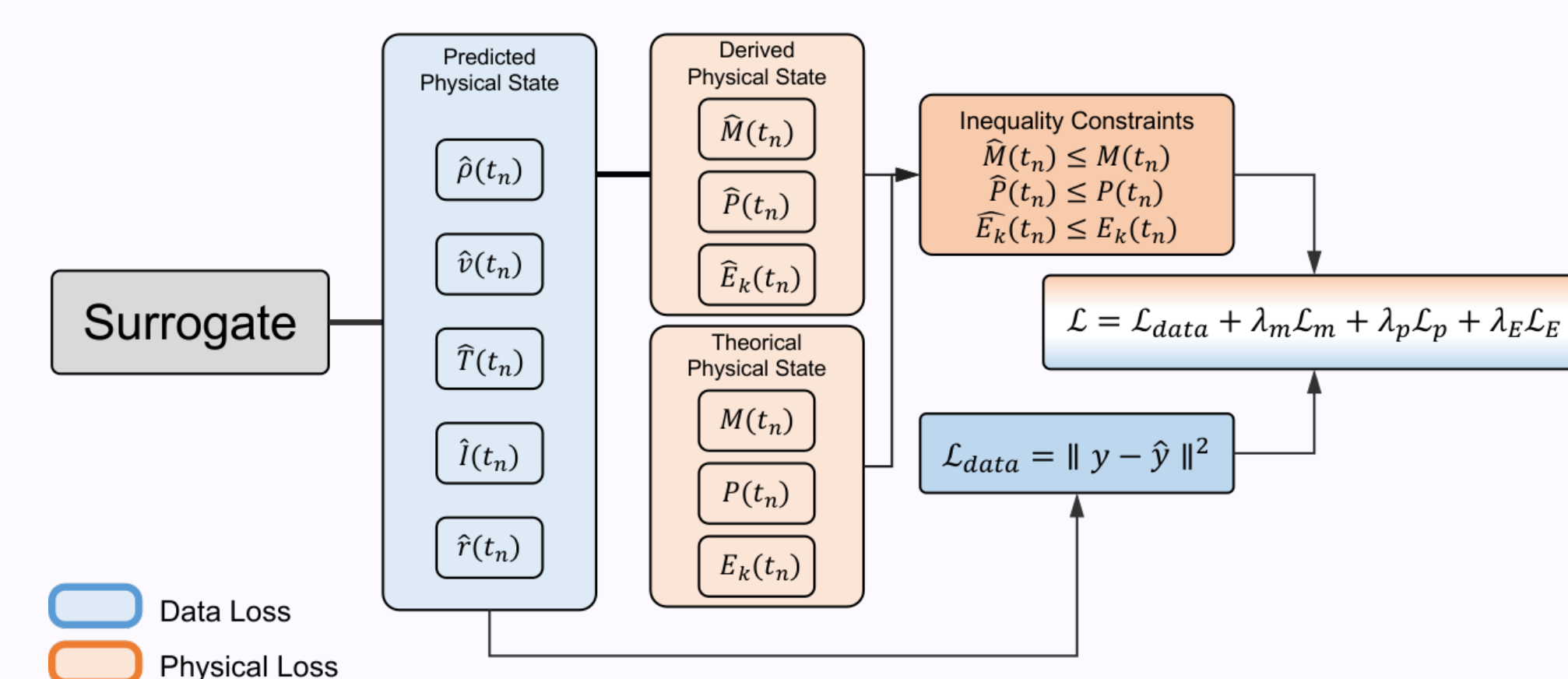
VAE, PPCA, and Fourier augmentation expand 701 measured pulses into a broader distribution of diverse pulse shapes.

■ **~90k pairs:** 701 Real, 10k Clean, 40k PPCA, and 40k Fourier OOD.

■ **I/O:** 160-step pulse + 3 targets → 5 trajectories × 160 steps.

■ **Leakage-safe split:** split by Shot ID before augmentation; held-out shots are never augmented; Fourier forms the OOD test set.

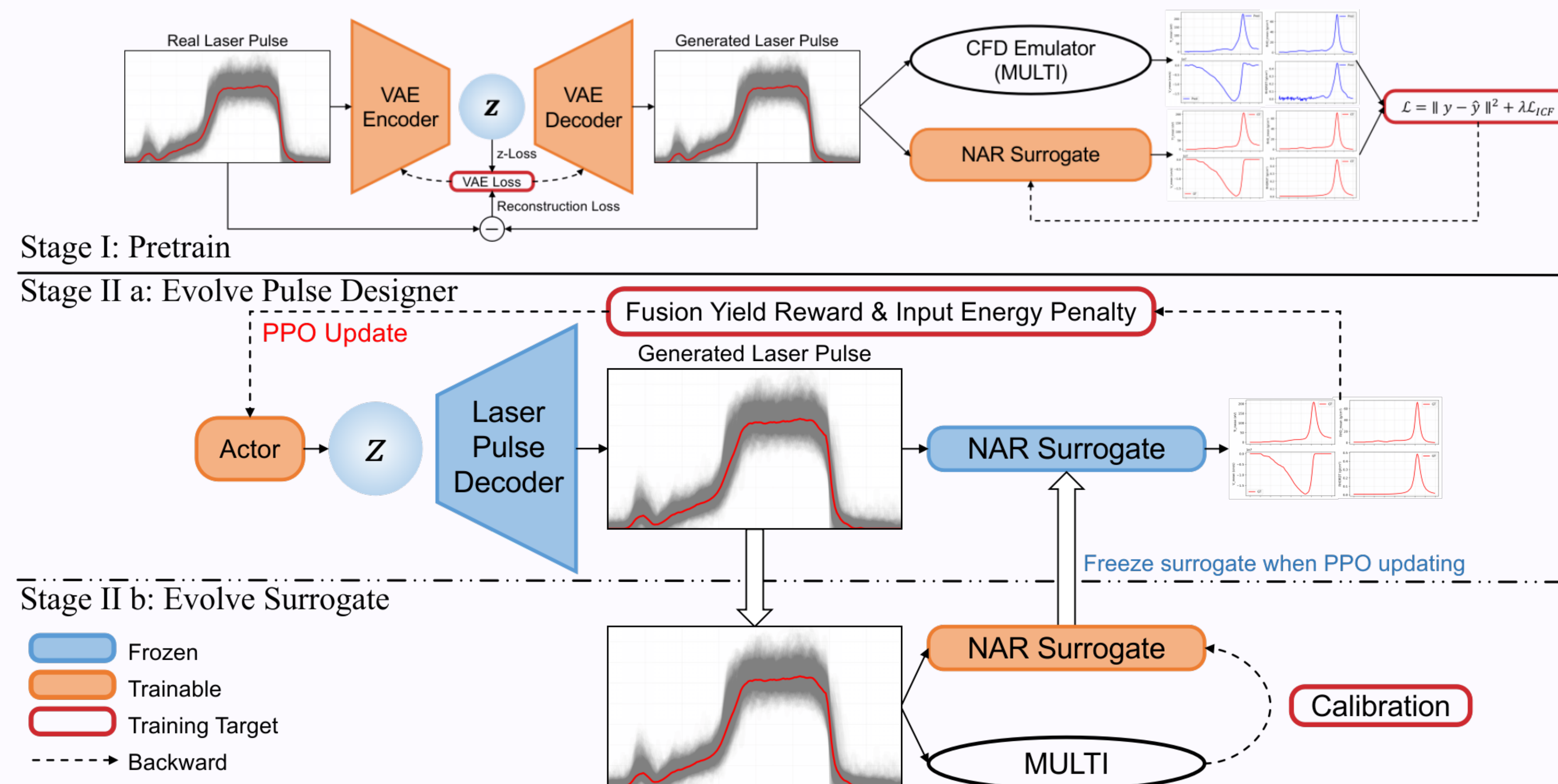
Physics as guardrails



One-sided penalties enforce mass, momentum, and kinetic-energy consistency, steering the surrogate toward physically valid trajectories.

Transformer encoder · Fourier OOD		
Metric	w/o physics	w/ physics
OOD Pearson ↑	0.855	0.912
Peak- x MAE ↓	8.020	6.302
Peak- y MAE ↓	0.310	0.252

Dual-frequency co-evolution

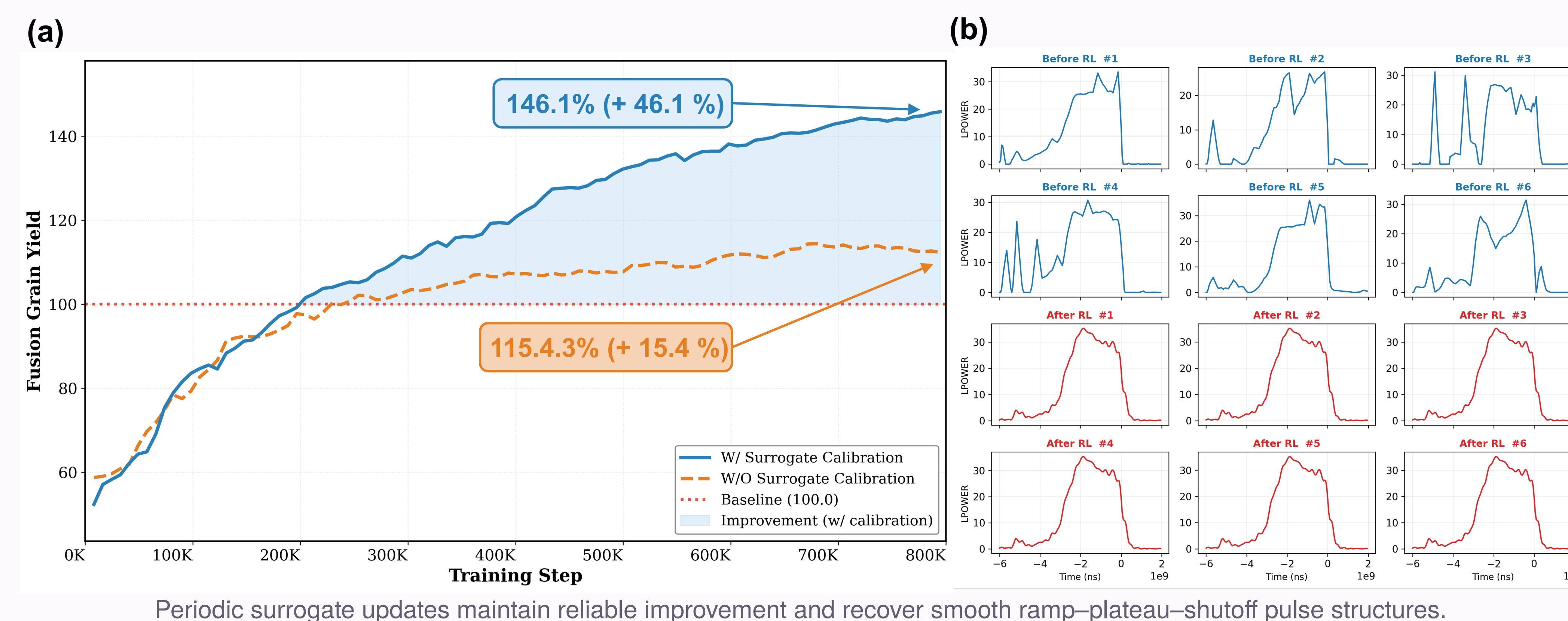


• **Stage I — pretrain:** learn a feasible pulse manifold and a physics-informed surrogate.

• **Stage II — co-evolve:** PPO searches on S_θ ; every K updates, MULTI labels current-policy pulses and refreshes S_θ on $\mathcal{D}_0 \cup \mathcal{D}_{\text{on}}$.

• **Key idea:** treat optimizer-induced shift as on-policy training data.

Calibration prevents reward hacking



Periodic surrogate updates maintain reliable improvement and recover smooth ramp–plateau–shutoff pulse structures.

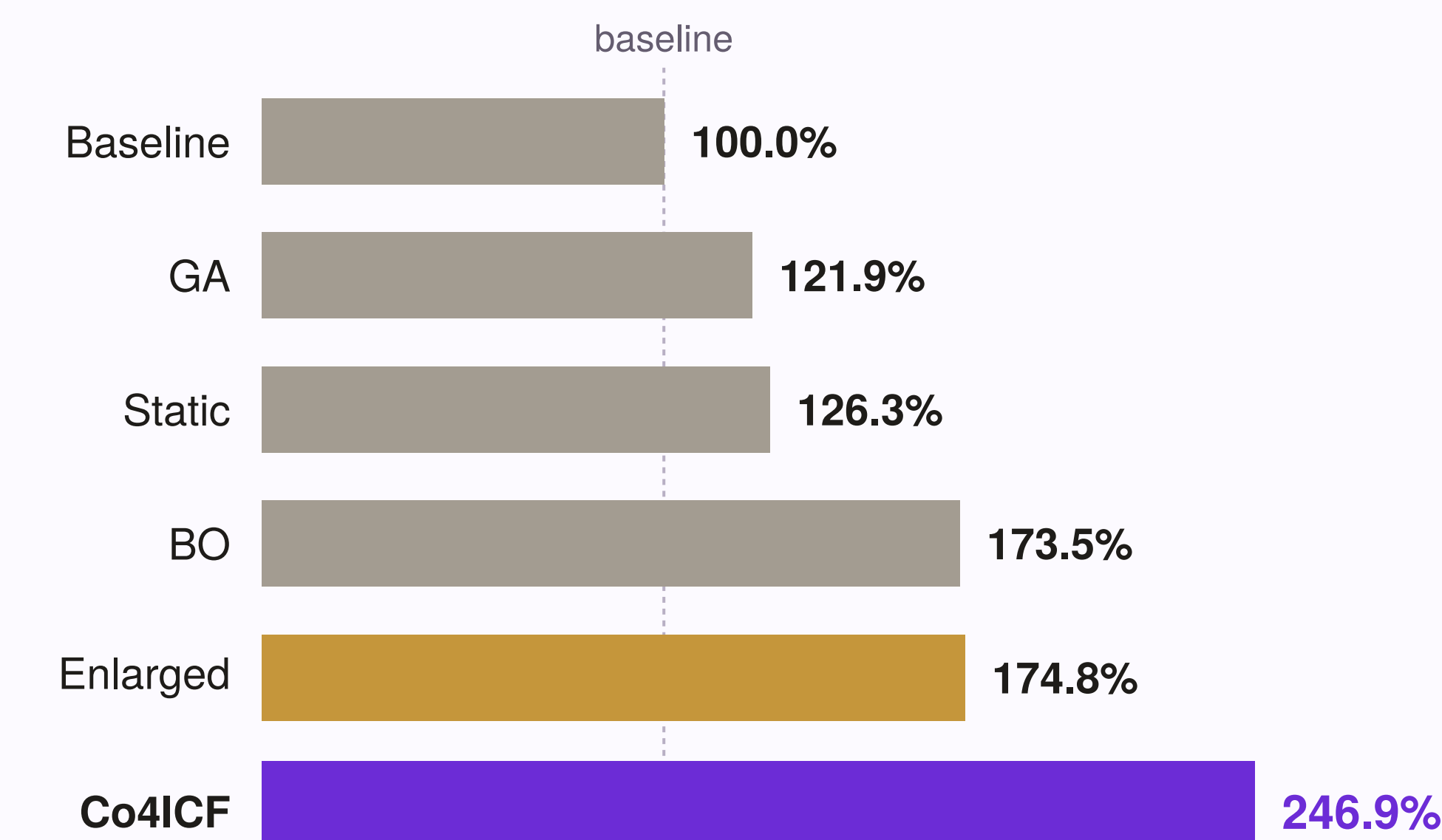
• **Normalized yield (1D-MULTI):** **146.1%** vs. 115.4% for the co-evolving and frozen surrogates.

• **PPO rollout time:** **20 s** with the surrogate vs. **19,800 s** with MULTI across 3,967 iterations (**990× faster**).

• **Timing scope:** PPO updates only; excludes pretraining, relabeling, and 2D verification.

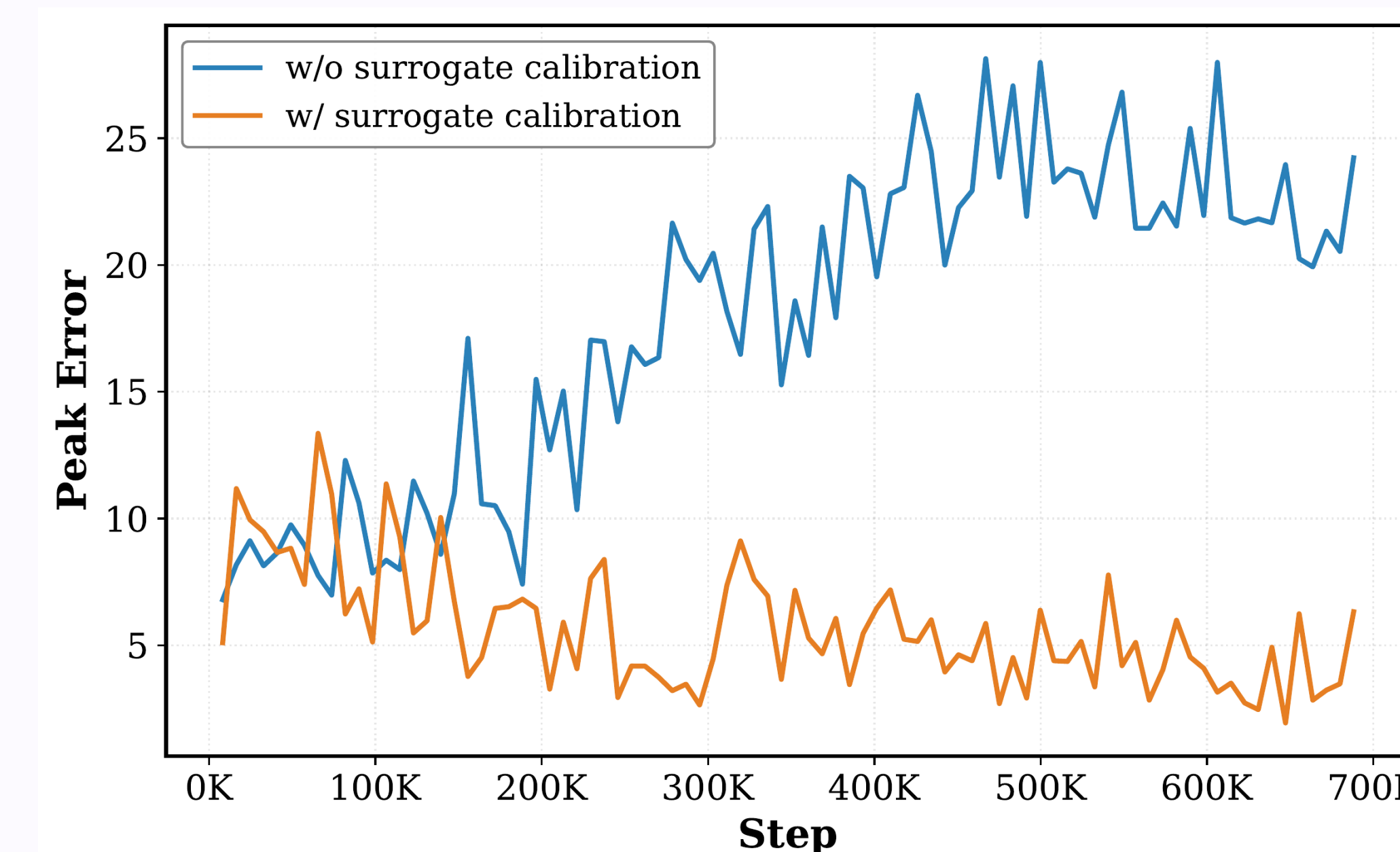
Direct 2D-MULTI result

Every method uses the same 1D backend and MULTI labeling budget. 2D-MULTI only evaluates final designs—never trains, tunes, or selects them.



Enlarged uses the same number of extra samples, but draws them from the original distribution. The budget-matched gap supports **on-policy** placement—not sample count alone.

Stay calibrated as policy shifts



Frozen-surrogate error grows as PPO moves OOD; periodic MULTI calibration keeps the surrogate–simulator gap bounded.

Takeaway

Co-evolution turns optimizer-induced distribution shift into useful training signal.

• **Stay reliable.** MULTI relabels current-policy pulses to keep the surrogate calibrated where PPO searches.

• **Search efficiently.** **20 s** surrogate PPO rollouts vs. **19,800 s** with MULTI across 3,967 iterations (**990× faster**).

• **Verify the gain.** Direct 2D-MULTI reaches **246.9%** vs. **174.8%** for the best budget-matched comparator.

Spend high-fidelity labels where the optimizer goes.

Code: github.com/Co4ICF/co4icf
Data: huggingface.co/datasets/Oyhs/Co4ICFDataset

